Structural insights into the inhibition of the drug efflux activity of the Hedgehog receptor PATCHED1

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ABSTRACT

PATCHED1 is the receptor of the HEDGEHOG morphogen and transports cholesterol in a proton motive force dependent way. In cancer, it is responsible for chemotherapy drug efflux. Screening of a natural product bank identified panicein-A hydroquinone (PAH), a marine sponge compound, as an inhibitor of PATCHED1-mediated dxr resistance, highlighting its potential as a therapeutic target [1,2].

In order to better understand the efflux inhibition activity of PAH towards PTCH1, we over-expressed and purified PATCHED1 in HEK293 cells, measured its affinity for PAH, and solved its structure in complex with PAH by cryo-EM. We identified a PAH binding site in the extracellular region and the residues involved in this binding pocket. PAH binds in the channel linking the sterol sensing domain to the extracellular medium (Fig 1A). It makes hydrogen bonds with the protein, either directly or via water molecules (Fig 1B) and leaves a large cavity empty (Fig 1 C), opening the way to structure-based drug design to improve the drug potency. Molecular dynamics (MD) simulations showed that PAH is stable in the binding site and also makes strong. These results, deposited in a Biorxiv preprint, provide key information on the mechanism by which PAH inhibits PTCH1 efflux activity [3].

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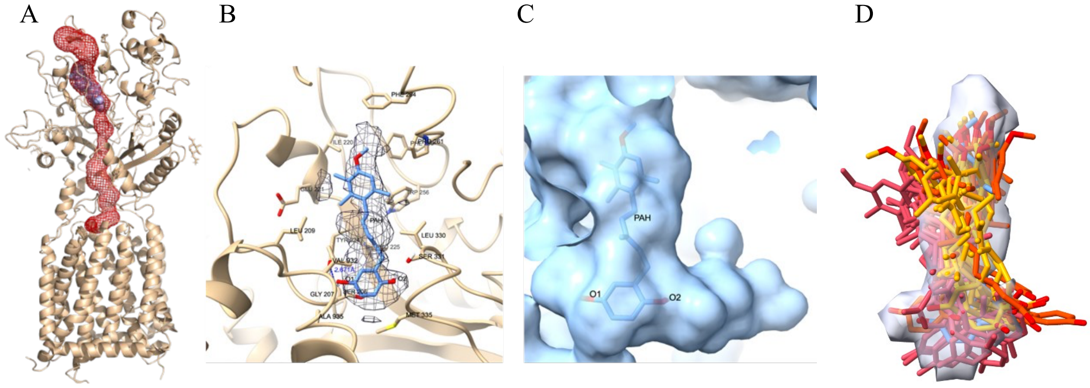

Figure 1 Cryo-EM structure of PAH-bound Patched1 receptor. A) PAH binds in the efflux channel linking the sterol sensing domain and the extracellular medium. B) cryo-EM density map and model of PAH with surrounding residues. C) protein surface in the PAH region shows a large cavity allowing for drug improvement. D) superposition of the MD simulations (yellow, orange, red) and of the cryo-EM map depicted as a white volume, shows that the MD confirms PAH location and interaction with the protein.

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